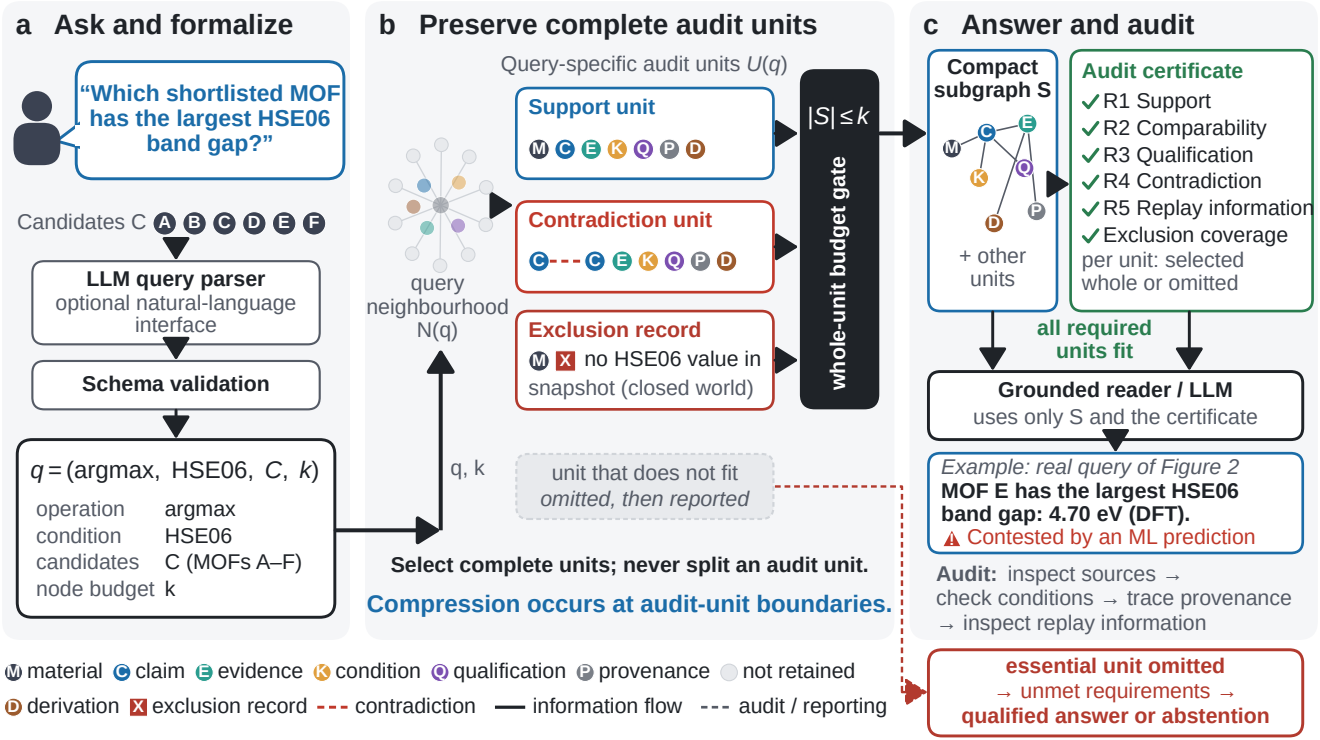

a Ask and formalize
"Which shortlisted MOF has the largest HSE06 band gap?"
Candidates C A B C D E F
LLM query parser
optional natural-language interface
Schema validation
q = (argmax, HSE06, C, k)
operation argmax
condition HSE06
candidates C (MOFs A–F)
node budget k
b Preserve complete audit units
Query-specific audit units U(q)
query neighbourhood N(q)
Support unit
M C E K Q P D
Contradiction unit
C --- C E K Q P D
Exclusion record
M X no HSE06 value in snapshot (closed world)
|S| ≤ k
whole-unit budget gate
q, k
unit that does not fit
omitted, then reported
Select complete units; never split an audit unit.
Compression occurs at audit-unit boundaries.
c Answer and audit
Compact subgraph S
+ other units
Audit certificate
R1 Support
R2 Comparability
R3 Qualification
R4 Contradiction
R5 Replay information
Exclusion coverage
per unit: selected whole or omitted
all required units fit
Grounded reader / LLM
uses only S and the certificate
Example: real query of Figure 2
MOF E has the largest HSE06 band gap: 4.70 eV (DFT).
Contested by an ML prediction
Audit: inspect sources → check conditions → trace provenance → inspect replay information
essential unit omitted
→ unmet requirements →
qualified answer or abstention
M material C claim E evidence K condition Q qualification P provenance not retained
D derivation X exclusion record --- contradiction — information flow --- audit / reporting